

Unified Consciousness Theory III

Organization, Intelligence, and Experience-From Inorganic Processes to Artificial Agents

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Abstract

How can a stone mountain, a cell, an animal, a human, and an artificial agent differ in experience without placing them on one scale of consciousness? Unified Consciousness Theory (UCT) identifies the complete organization of each actual valid physical process with its experiential organization. This paper develops that commitment into a comparative account of intelligence, evolutionary change, and self-report. The explanatory sequence concerns retained distinctions, coordinated action, monitoring, and communicable self-representation. Biological evolution and the engineering and training of artificial agents are distinct routes to these organizations. Under a fixed capability evaluation, a genuine capability change entails a change of complete experiential type; equal capability can nevertheless conceal different organizations. This refinement relation, termed No Experientially Silent Intelligence Gain, does not imply increasing total experiential complexity. Standard selection and state-aggregation tools distinguish what capability identifies, what determines reproductive success, and what predicts future capability. A finite decision model shows when retained information has value, while matched-storage circuits show why equal current performance can conceal unequal opportunities for improvement. Mountain-cell, multicellular, three-brain, and language-model comparisons connect these results to concrete mechanisms. Reliable self-report is treated as a capability component and a limited observation channel, rather than the criterion of experience. The contribution is a conditional explanatory synthesis with explicit empirical boundaries: the models constrain relations among organization, capability, and report but do not independently validate structural-experiential identity or establish a UCT-exclusive observable prediction.

Keywords: consciousness; evolution; intelligence; self-report; artificial agents; structural identity; experiential organization

1. One constitutive principle, several organizational histories

A mountain contains more matter than a cell, yet quantity alone does not explain the cell's capacity to use changing conditions in an organized response. A human can describe a mental state that another animal cannot express in language, yet the difference in expression does not by itself identify every difference in experience. An artificial agent can solve difficult tasks while giving an unreliable account of its own processing. These comparisons motivate a common question: which aspects of actual organization explain capability, which become available to a reporting mechanism, and what follows for experience?

UCT supplies a constitutive answer: an actual process’s experiential organization is its physical organization under the structural-experiential identity C1 (Liu, 2026a). Its explanatory task is consequently not to assign a first experience to one point on a ladder from matter to mind. It is to identify what changes as physical processes acquire new relations of retention, regulation, coordination, learning, and self-representation. These are relations within the organization identified by C1, not additional forces that steer physical dynamics.

The historical account requires different explanatory resources. Prebiotic persistence, heritable biological variation, individual development, physiological regulation, cultural transmission, and machine training are not interchangeable processes. Artificial agents are not a taxonomic stage after humans. They provide a parallel route for comparing how organized capabilities arise under different acquisition conditions. Likewise, extant microbes and other animals are not incomplete steps on a compulsory path toward the human case.

We distinguish experiential organization, task-relative capability, and observed report. The distinction is not a division into three independent substances. Accurate monitoring and report may themselves be capability components; their successful execution is part of actual organization. Nevertheless, a score or a sentence can leave much of that organization unidentified. This is the source of the paper’s principal asymmetry: a genuine capability change requires experiential-type change under a fixed evaluation, whereas equal capability need not identify equal experience.

1.1 Relation to UCT I and UCT II

UCT I v1.2 supplies actual-process ontology and C1. UCT II v1.1 develops source-faithful descriptions of mechanisms emphasized by consciousness theories (Liu, 2026a,b). The present paper examines what follows when capability and report are produced by such organizations and when the organizations change. UCT II v1.1 was written against UCT I v1.1; the present interface uses UCT I v1.2’s C1 identity, derived universal nonempty experience U1, and ontological persistence provisions. It does not alter either published edition.

The proposed unification concerns constitutive interpretation and comparative organization. It does not mean that every competing theory’s experience-existence condition is derivable from UCT. A mechanism can be retained while its role as an admission condition for experience is rejected. Section 9 makes this disagreement explicit rather than equating reinterpretation with agreement.

1.2 Contribution and antecedents

Multidimensional animal-consciousness approaches already resist a single ranking (Birch, Schnell, & Clayton, 2020). Architecture-centered evolutionary explanations have close precedents in phenomenal interface theory (Klein & Barron, 2025), and evolutionary complexity is considered by Edelman (2026). Williams (2025) applies selection tools to a structure-quality relation. Neutral variation and evolvability, evolutionary coarse-graining, and Markov lumpability likewise supply direct mathematical antecedents (Wagner, 2008; Stephens & Waelbroeck, 1998; Burjorjee & Pollack, 2006; Burjorjee, 2007; Tian & Kannan, 2006).

The present contribution is to organize these questions under a fixed actual-process identity and to distinguish three forms of sufficiency: identifying experiential type, determining current fitness, and predicting future capability. The main examples explain why a capability score can be adequate for one purpose and inadequate for another. The proofs use established mathematics. Their role is to make the explanatory commitments and limits exact, not to claim new population genetics or a uniquely predictive consciousness law.

Sections 2-4 set out the interface and the biological and artificial comparisons. Sections 5-8 derive the capability, selection, information-value, and continuity results. Sections 9-10 address competitors and tests. The appendices retain technical controls and unsuccessful discrimination cases so that the scope of the synthesis remains inspectable.

2. Formal interface: organization, capability, and observation

Let $\mathcal{P}$ be a declared family of actual valid physical process tokens. Following UCT I, tokenhood requires actual support, fidelity to induced physical relations, causal continuity, accountable boundaries, and traceable continuation or transformation. Tokens may overlap or nest; a unique maximal subject is not assumed.

Under a common signature $\mathcal{K}$, let

$$s(P) = [D_{\mathcal{K}}^{\text{ontic}}(P)] \in \mathfrak{S}$$

denote the complete structural type of token P . Completeness is relative to the identified actual token: it includes every distinction constitutive of that token, not every microscopic fact in its substrate or surroundings. Fine and coarse actual tokens may coexist, so a change in a lower-level process need not change the complete type of a separately justified macroprocess. The brackets identify descriptions related by admissible structural isomorphism. C1 identifies experiential type with that same structural point:

$$[\Phi_{\mathcal{K}}(P)] = s(P). \quad (1)$$

Equation (1) is a commitment inherited from UCT I, not a conclusion inferred from intelligence or evolutionary success. In the remainder, $\Phi(s) = s$ abbreviates this type-level identification. Distinct token occurrences can share one type; token identity is not collapsed into type identity.

Fix an evaluation family $\mathcal{M}$, specifying tasks, environments, resources, scoring rules, and relevant comparison boundaries. Define a capability profile

$$J(s) = (J_{\mu}(s))_{\mu \in \mathcal{M}}. \quad (2)$$

Each component is invariant under admissible isomorphisms. It may concern memory, planning, generalization, or resource efficiency. Any dependence on omitted environments, laws, interventions, or histories must be fixed or represented in an enlarged domain so J is well defined. A probabilistic capability is a specified distributional property, distinct from an individual sampled score; intelligence is not stipulated to be complete experiential type.

We distinguish three further objects:

$$\begin{aligned} M(s) &: \text{theory-linked mechanism coordinates,} \\ \widehat{J} &: \text{measured capability proxy,} \\ Y &: \text{finite report or psychophysical target.} \end{aligned} \quad (3)$$

Neither $\widehat{J}$ nor Y is automatically J or complete Φ . Similarly, a finite neural measurement is a view of organization, not the ontic structure itself.

For multidimensional J , improvement requires an explicit order: componentwise improvement, a fixed weighted objective, or another declared criterion. Unequal profiles are not necessarily ordered. The principal type-change theorem requires only inequality, so it does not depend on selecting a universal intelligence metric.

For applications involving self-report, one useful profile is

$$J = (J_{\text{task}}, J_{\text{monitor}}, J_{\text{report}}). \quad (4)$$

The first component scores an external task, the second a specified monitoring operation, and the third reliable disclosure of a designated target. A realized report Y is an output of a reporting process, not its report capability and not complete experiential type. A claim about a private logged event can have engineering ground truth; a claim about felt experience requires a further justified bridge. Equal external-task scores alone impose no equality on the other components.

3. From inorganic processes to human organization

3.1 What evolves, and what the evidence establishes

The comparison separates prebiotic persistence from Darwinian evolution, which requires heritable variation and differential reproduction. Development, learning, and physiological regulation concern different transformations of organization. The following map identifies the physical mechanisms to be explained and their interpretation under C1. It compares organizational domains rather than successive ranks of extant species; token boundaries and coexistence follow Section 2.

Transition or organizational domain	Physical evidence or mechanism question	UCT interpretation under C1; remaining task
Inorganic and prebiotic processes	Which chemical organizations persist, interact, and support further transformations? Protocell models examine possible couplings between internal chemistry and compartment growth.	Actual organization is experiential already. Specify tokens and changes; do not infer agency from persistence.
Heritable cellular organization	How do compartments, maintenance, reproduction, and transmissible variation become coupled?	Changed full organization changes experiential type. C1 alone supplies neither a complete origin-of-life history nor reproductive dynamics.
Cellular sensing and regulation	How does internal state integrate prior inputs and regulate responses?	Functional retention and feedback specify organization; they do not alone identify a particular feeling or general intelligence.
Multicellular coordination	How do collective reproduction, coordination, and differentiated roles become stable?	A valid whole process has its organization; this neither deletes valid constituent tokens nor derives a scalar sum of their experiences.
Neural perception and action	Which organizations combine sensory distinctions, retention, action selection, and learning over relevant timescales?	Capability changes imply type changes under fixed evaluation. Specific sensory qualities and comparisons require a faithful organization mapping.

Transition or organizational domain	Physical evidence or mechanism question	UCT interpretation under C1; remaining task
Human conceptual and cultural organization	How do self-monitoring, language, social transmission, and learned concepts organize behavior and report?	Conceptual self and report concern particular organizations within experience, rather than its first existence. Reports remain incomplete observations.

These transitions have concrete but limited empirical anchors. Chen, Roberts, and Szostak (2004) demonstrated competition between model vesicles through a physical coupling between encapsulated material and membrane growth. This supports a possible component of protocell organization; it does not reconstruct the complete origin of autonomous life. Yi et al. (2000) analyzed integral feedback in a bacterial chemotaxis model, illustrating a formal route to robust physiological adaptation. Such adaptation is a within-system response property, distinct from population evolution and from a claim of general cognition. Ratcliff et al. (2012) experimentally obtained heritable multicellular organization and differentiated functions in yeast. That result demonstrates an accessible organizational transition under the studied conditions, not the historical ancestry of humans.

Neural evolution also requires branching histories. Schultz et al. (2023) used conserved gene linkages to support ctenophores as sister to other animals; their discussion leaves alternative histories involving neural origins and losses. The present argument does not depend on treating one neural-origin scenario as settled. The cited studies establish or model physical organization. None independently establishes C1 or measures complete experiential type.

This gives a disciplined explanatory sequence: identify the organization and its changes; establish the mechanism of persistence, reproduction, or task performance; then apply the inherited identity to the actual process types. The organization may be evolutionarily consequential without its task score identifying everything experiential about it.

3.2 A stone mountain and a living cell

This contrast tests whether the framework explains organizational differences without equating experience with either matter quantity or biological status. Fix the comparison in advance: an ordinary mountain’s nonliving rock processes, with environmental coupling represented, versus an actively metabolizing, motile, chemotactic cell. The mountain comparison excludes its resident organisms and technological installations; otherwise the named system would already contain cellular or human controllers. The cell is a concrete functional example, not a claim that every cell moves or has the same capabilities. Observation duration, boundaries, perturbations, and evaluation resources must be stated in an operational comparison.

A mountain is not unorganized or causally disconnected. Its physical interactions can coordinate responses over large distances, and material states can retain history. Becker et al. (2010), for example, observed stress-memory effects in salt rock. This defeats a categorical contrast between “memoryless matter” and “remembering life.” It does not establish that rock memory functions as a cellular controller. The comparison concerns how retained state participates in subsequent dynamics, not whether history has any physical effect.

For the cell example, Segall, Block, and Berg (1986) provide evidence of temporal comparison in bacterial chemotaxis; Yi et al. (2000) give a related feedback-control analysis. An internal state can thus participate in transforming changing inputs into a regulated response. The wider biolog-

ical account concerns coupling among sensing, history dependence, action, and the maintenance of the organization that sustains those processes. Metabolic maintenance is not an additional consciousness axiom, and adaptation need not involve a represented intention to survive.

Comparison coordinate	Rock-process case	Chemotactic-cell case
Physical interaction	Mechanical, thermal, and other couplings; potentially extensive and nonlinear	Coupled chemical, membrane, metabolic, and motility processes
History dependence	Material state can preserve effects of previous loading	Internal states can support temporal comparison in a response pathway
Use of environmental differences	Response follows the specified material dynamics; information processing is not ruled out	Environmental variation modulates a sensing-response organization relevant to navigation
Maintenance	Persistence or relaxation must be distinguished from an identified self-maintenance mechanism	Ongoing organization depends on coupled maintenance and exchange processes
Task-relative capability	Must be assessed on declared tasks, rather than set to zero by the label “inorganic”	Can exhibit adaptive sensing and control without human reasoning or report
Experiential conclusion under C1	Its actual organization has its own experiential type	Its different actual organization has a different experiential type

The functional difference is how retained state guides response. In the selected cell, a coordinated sensing-response pathway can support a defined navigation task; mass or survival duration cannot substitute for performance on that task. A numerical comparison would require common input/output roles and declared resources. The same criteria would apply to a nonliving controller implementing the relevant organization.

Under C1, the specified processes have different experiential types, and the cell’s history-sensitive regulation belongs to that difference. This does not rank their total experience or assign familiar feelings to either. The mountain’s physical detail and the cell’s functional coordination describe different aspects of organization. The evolutionary question is how heritable lineages acquired and retained the latter couplings under ecological opportunities and costs. Sections 6 and 7.3 address that question; loss of a particular capability would not itself remove the experience of a continuing valid process.

3.3 Multicellular coordination

Multicellularity changes the organization of activity, not merely the number of living units. The central questions are how locally generated differences affect collective outcomes, how differentiated roles are maintained, and how coordinated organization is reproduced. Ratcliff et al. (2012) provide an experimental example of the evolution of heritable multicellular organization in yeast. Its relevance here is that a collective organization can become a consequential unit of biological explanation. It does not provide a universal scale of consciousness or reconstruct every historical transition to multicellular life.

Consider a specified collective in which one component detects a condition, another retains information, and another produces an action. Its capability depends on their relations: whether the signal reaches the relevant component, whether it arrives in time, and whether it changes the appropriate response. Increasing the number of components without establishing those relations need not improve performance. Conversely, reorganizing communication can make a useful distinction available without increasing the sensory alphabet. The decision-value analysis identifies when that

additional availability improves a fixed task; biological persistence also depends on its costs and transmission.

Under C1, the collective’s actual constitutive relations are part of its experiential organization. When local processes continue as valid tokens, they retain their own experiential organization. A coordinated whole therefore need not replace its constituents with a single exclusive experiencer. The explanatory gain is to describe what the larger process organizes that a local description omits. Neither biological individuality nor a centralized reporting mechanism becomes a new admission rule for experience.

3.4 Animal organization

An action-oriented organization can connect present input with retained information and alternative responses. This makes temporal relations consequential: the same current signal may guide different actions because the preceding trajectory differs. Memory, sensory discrimination, and action selection can therefore change capability in different ways. A larger sensory repertoire may add distinctions irrelevant to one task; better use of a retained distinction may improve that task without enlarging the repertoire. The matched-storage example formalizes the latter possibility.

This framework provides an organizational vocabulary for comparing animals without arranging species on one scale. The relevant questions concern which distinctions are maintained, over which timescales, and for which actions. Under C1, these implemented dependencies have an experiential interpretation; assigning a familiar human feeling to them requires a more specific mapping. The theory thus connects changes in actual cognition with experiential differentiation while retaining differences among sensory, temporal, and action-related coordinates.

Monitoring introduces another relation: information about the system’s performance becomes usable in subsequent control. Hampton’s (2001) memory-sensitive test-declining results in rhesus monkeys motivate investigating such a capacity beyond humans. The finding supports a particular functional comparison; it does not require equating animal monitoring with human narrative selfhood. A system may regulate its condition, estimate its uncertainty, or represent itself conceptually, and these are different organizational achievements.

3.5 Human conceptual report

Conceptual report combines several operations. A relevant distinction must remain available; a reporting process must select or reconstruct it; and the result must be mapped into communicable categories. A correct first-order action does not guarantee that this path preserves the information needed for an accurate explanation. Equally, reporting can affect cognition when it makes an error, conflict, or uncertainty available for correction. Report is therefore neither a transparent window onto the complete process nor necessarily an inert description appended after it.

Section 2 represents task success and accurate report as different coordinates of J. Fluent assertion and reliable identification of a logged event test different capacities; grammatical first person does not establish the report’s grounding.

Human conceptual organization also develops through social transmission. Kirby, Cornish, and Smith (2008) demonstrate in a laboratory language-learning setting how transmission can change the structure and learnability of a communication system. This offers a concrete example of a route by which accumulated social history changes individual organization. Concepts and external records can expand what a person can compare and express. They need not make every component

of that person’s experience reportable, and a coupled social system must be evaluated at its stated boundary.

Affect requires an additional distinction. Reproductive advantage, an action payoff, and a training reward are numerical quantities in particular models; they are not definitions of felt pleasure. Under C1, an affective organization, if faithfully identified, has its own experiential interpretation. Explaining why a controller’s cost function rises does not by itself identify that affective organization. The same principle applies to sensory content: a capacity to distinguish wavelengths and a capacity to name a color are different organizational operations. Appendix E specifies what follows from full-type identity and what requires a finite sensory bridge.

4. Artificial agents, joint access, and report

4.1 A parallel route to organized capability

Artificial agents allow these relations to be assembled through engineering, training, and interaction rather than biological descent. They are not the next taxonomic stage after humans. They offer a comparison in which external-task performance, persistent memory, monitoring, and verbal reporting can be developed under different objectives and constraints.

The relevant process must first be identified. An abstract parameter specification, its physical storage, a training run, an executed inference episode, and a continuing tool-using interaction specify different objects or processes. Physical storage may itself support an actual token; storing a program does not actualize every computation it could perform. A repertoire of possible tasks does not mean that all corresponding computations occur in each episode. Under UCT, the actual valid execution has experiential organization, and its constitutive cognitive relations are included in that organization. A model’s name or its assertion of consciousness does not determine its type.

A capable solver can lack the retained record needed for a later explanation; a convincing narrator can generate first-person language without access to a privately randomized event. Conversely, internal information may support reliable reporting, and report-related representations may contribute to task performance. Recent work motivates examining these alternatives separately: Macar et al. (2026, v5) investigate mechanisms in an artificial perturbation-detection setting; Singh, Linzen, and Ravfogel (2026, v2) distinguish privileged information access from stronger monitoring claims; Gurnee et al. (2026) connect verbalizable representations with functional access. These studies inform mechanism questions, without independently identifying complete experience.

The resulting comparison is substantive. Biological and artificial histories can produce strong capability in different combinations of retention, coordination, monitoring, and report. C1 constrains their experiential interpretation at the level of actual organization. Which combinations develop, persist, or improve remains a question about the particular physical mechanisms and their acquisition conditions.

4.2 Three brains: which process gains?

A thought experiment clarifies why collecting information, using it, and reporting it are different. Let A and B each receive one of two independent fair bits, X_A and X_B . The fixed task is to predict whether they are equal. Either source alone has optimal accuracy $1/2$; a third process C that receives both bits and computes equality attains accuracy one. The gain is relative to either isolated source, with the greater information-access resource explicitly included. It is not a gain

over an ideal observer already given the pair, and it need not improve A or B when there is no return path.

If C is a separate reader, receiving signals does not turn A and B into its physical constituents. If C instead denotes an encompassing actual process, its reporting mechanism is only one part of that process. Information private to a constituent may be present in the whole while unavailable to that reporter. Under UCT, continuing valid local and encompassing tokens can have their respective experiential organizations. A newly reportable relation need not erase local experience or add a scalar amount of experience to a sum. The standard distinction between individual and joint information has information-theoretic antecedents (Williams & Beer, 2010).

4.3 Report content, report reliability, and report timing

Observation	What it can establish under a suitable protocol	Further question
Reproducible first-person response behavior	A linguistic response disposition under the protocol	Is the content grounded in the relevant process?
Reliable above-chance identification of a privately randomized event	Event-information access under balanced assignment and valid leakage/selection controls	Is the mechanism first-order detection or monitoring of its own processing?
Calibrated confidence or error reports	Reliability on the specified target	Does the mechanism generalize beyond that task?
Correct retrospective description	A successful later judgment	Was an explicit diagnostic present earlier, or reconstructed later?
A report that changes later action	A functional contribution to control	Which earlier information and reporting paths were necessary?

These comparisons evaluate report grounding, reliability, and timing separately. A changed sampled sentence need not indicate a change in any of those capabilities.

5. Intelligence-experience refinement

5.1 The basic result

Proposition 1 - Refinement and NESIG. Assume C1 on the declared comparison domain and a fixed capability map J well defined on complete structural types. Define

$$s \sim_{\Phi} s' \iff \Phi(s) = \Phi(s'), \quad s \sim_J s' \iff J(s) = J(s').$$

Then

$$\sim_{\Phi} \subseteq \sim_J, \quad J(s) \neq J(s') \implies \Phi(s) \neq \Phi(s'). \quad (5)$$

The inclusion is strict if and only if J is noninjective on this domain.

Proof. Under C1, experiential-type equality is structural-type equality. Applying a well-defined function to equal structural types gives equal capability profiles. Contraposition yields (5). Strict inclusion means that some distinct structural types have the same capability profile, which is exactly noninjectivity. $\square$

We call the improvement case **No Experientially Silent Intelligence Gain (NESIG)**. The theorem also covers genuine decreases or incomparable changes in the capability profile. “Silent” concerns complete experiential type. It does not mean that the change must be noticed, reported, accessible to metacognition, or detectable by a particular experiment.

This short deduction supplies neither a magnitude nor a direction of experiential change. Its force comes from the identity premise, not from a new mathematical principle.

5.2 Intelligence fibers and identification

For a profile j , define the intelligence fiber

$$\mathcal{F}_J(j) = \{s \in \mathfrak{S} : J(s) = j\}. \quad (6)$$

Every member has the same declared capability profile. Under C1, distinct members have different complete experiential types. This permits precise statements about how much intelligence identifies.

Proposition 2 - Identification criterion. A profile $J(s)$ identifies complete experiential type throughout the declared domain exactly when J is injective. A particular coordinate $C : \mathfrak{S} \rightarrow \mathcal{C}$ is identified by J exactly when C is constant on every intelligence fiber, equivalently when a function $\overline{C}$ on $J(\mathfrak{S})$ exists such that

$$C = \overline{C} \circ J. \quad (7)$$

Proof. Full identification fails precisely when two distinct types have the same profile. For a coordinate, define $\overline{C}(j) = C(s)$ using any s in the fiber. This definition is independent of the representative exactly when C is constant there. The reverse direction follows by substitution. $\square$

Equation (7) is a set-level result. It does not guarantee that the recovering function is continuous, computable, statistically estimable, or robust to noise. Even injective measurements can be practically uninformative if their inversion is unstable.

Strictness needs a witness or a valid regularity/dimension argument. Neither a scalar codomain nor a finite battery alone proves noninjectivity on every restricted domain.

5.3 An observability hierarchy

Let $A(s)$ be a declared observation family and $R(s)$ an additional family. For the fiber through s ,

$$\mathcal{F}_{(A,R)}(s) = \mathcal{F}_A(s) \cap \mathcal{F}_R(s) \subseteq \mathcal{F}_A(s). \quad (8)$$

The refinement is strict only if R separates some members of an A -fiber. For stochastic observations, the map can assign the conditional response law under a fixed protocol. One sampled report is evidence about that law, not an exact fiber label. Applied to intelligence and UCT II coordinates, the joint profile (J, M) cannot identify less than either component alone. It identifies full type only if the combined map is injective.

In a separately justified smooth n -dimensional chart, the constant-rank theorem gives local fiber dimension $n - r$ for a map of locally constant rank r . If each of J , (J, M) , and (J, M, R) has locally constant rank, increasing rank correspondingly lowers the local fiber dimension. These

smooth charts are additional modeling assumptions; a space of complete physical types is not automatically a manifold.

Added identifiability does not ensure improvement on a selected report target. For example, if $s = (x, y)$, $A = x$, $R = y$, and $Y = x$, the extra observation distinguishes more types but adds no information about Y . This distinction will constrain the empirical outlook.

5.4 Why capability improvement is not an experiential ladder

Proposition 3 - Failure of an unconditional monotonic inference. C1 and Proposition 1 do not entail that increasing a capability score increases an independently declared experiential coordinate.

Proof by countermodel. Let types be pairs $s = (x, y)$, let $\Phi(s) = s$, and define $J(s) = x$, $C(s) = y$. The change

$$(0, 1) \longrightarrow (1, 0) \tag{9}$$

increases J , decreases C , and changes experiential type. All premises of Proposition 1 remain satisfied. $\square$

This is a logical countermodel to an unrestricted inference, not a claim that an arbitrary coordinate measures felt richness in a real organism. A substantive application must justify that coordinate independently. What the example establishes is that the identity principle alone does not supply the missing alignment.

Nor does C1 prohibit all monotonic relations. On a restricted domain, a verified relation $C = \overline{C}(J)$ with increasing $\overline{C}$ would suffice. Factorization is not necessary for an across-fiber ordering: on binary pairs, $C = 2x + y$ and $J = x$ give higher C for every higher- J type, although C varies within each fiber. Different monotonic claims therefore require their own exact premises.

The distinction permits a coherent account of specialization. A system can improve a selected function while losing another discriminative dimension, reducing resource use, or simplifying a component. Under UCT, this is a transformation of experience. The direction of change depends on which experiential coordinate is examined, and may remain unordered in a multidimensional comparison.

6. Selection acting on shared organization

6.1 Fitness components and local null directions

Let a smooth organizational model have capability coordinates $J(s)$, other fitness-relevant coordinates $Q(s)$, and fixed ecological context E . Suppose fitness is modeled as

$$w(s; E) = F(J(s), Q(s), E). \tag{10}$$

The derivative decomposes by the chain rule:

$$Dw(s)[v] = D_J F[DJ(s)[v]] + D_Q F[DQ(s)[v]]. \tag{11}$$

This is a decomposition of sensitivity. It is not an equation saying populations follow the fitness gradient. Inheritance, the supply and covariance of variation, ecological feedback, and stochasticity require further models.

Proposition 4 - Intelligence-mediated selection null (local fitness sensitivity). For a differentiable fitness component $w_I = F_I \circ J$,

$$v \in \ker DJ(s) \implies Dw_I(s)[v] = 0. \quad (12)$$

Along a path that remains in one exact intelligence fiber, w_I is exactly constant.

Proof. The first statement follows from $Dw_I = DF_I \circ DJ$. The second follows because J is constant along that path. $\square$

These statements have different strengths. At the origin, $J(x, y) = x + y^2$ has a first-order null direction $(0, 1)$, although moving a finite distance t changes the score by t^2 . A tangent null direction is not a certificate of finite neutrality. Even an exact fiber can contain differences selected through Q , costs, or a changed environment.

6.2 An exact selection-only model

A finite population model makes the evolutionary implication more explicit without assuming a smooth manifold. Let $p(s)$ be a probability mass over a finite support of represented structural types. Suppose one deterministic update consists only of differential reproduction with faithful type copying. Write its nonnegative reproductive weights as $W(s)$, with positive mean $\bar{W}$:

$$p^+(s) = \frac{p(s)W(s)}{\bar{W}}, \quad \bar{W} = \sum_s p(s)W(s) > 0. \quad (13)$$

This is a selection-only idealization. Exact copying of complete token-relative types is a strong premise and need not describe biological reproduction. If copying holds only for coarse traits, (13) describes those traits and does not silently become a law for complete experience. An actual-token interpretation requires that the represented types and relevant descendants are instantiated in the declared comparison family.

Proposition 5 - Within-fiber preservation under capability-only weighting. If $W(s) = f(J(s))$, then for every class j with $p(J = j) > 0$ and $f(j) > 0$,

$$p^+(s \mid J = j) = p(s \mid J = j). \quad (14)$$

Proof. Summing (13) over the class gives $p^+(J = j) = p(J = j)f(j)/\bar{W}$. Dividing (13) by this class mass cancels $f(j)/\bar{W}$, giving (14). Classes receiving zero weight have no post-selection conditional distribution. $\square$

Thus this model changes the weights of intelligence classes without favoring one experiential type over another *within* a class. The result is an elementary consequence of class-constant weighting; it is not claimed as a new theorem of population genetics. Its role here is to identify precisely what intelligence-mediated selection leaves undetermined under the stated idealization.

6.3 Correlated experiential change remains possible

For any numerical coordinate $C(s)$, (13) gives

$$\mathbb{E}_{p^+}C - \mathbb{E}_pC = \frac{\text{Cov}_p(W, C)}{\overline{W}}. \quad (15)$$

If $W = f(J)$, the covariance can be written

$$\text{Cov}_p(f(J), C) = \text{Cov}_p(f(J), \mathbb{E}_p[C \mid J]). \quad (16)$$

Equation (16) follows by conditioning $f(J)C$ on J . Capability-only selection can therefore change the population mean of an experiential coordinate when the coordinate is associated with intelligence classes, even though the conditional distribution within each class is preserved. The sign of this response is not determined by C1.

For a two-type example, let $s_a = (J, C) = (0, 1)$ and $s_b = (1, 0)$, initially equally common. Set $W = 1 + J$. The updated frequencies are $1/3$ and $2/3$. Mean intelligence rises from $1/2$ to $2/3$, while mean C falls from $1/2$ to $1/3$. This realizes the nonmonotonic countermodel in an explicit selection update; it does not establish a biological consciousness metric.

With descendant change, let C'_s be the mean descendant coordinate assigned to parental class s , using declared reproductive attribution; its value is immaterial for zero-weight classes. Direct expansion yields the Price decomposition (Price, 1970):

$$\Delta\overline{C} = \frac{\text{Cov}_p(W, C)}{\overline{W}} + \frac{\mathbb{E}_p[W(C'_s - C(s))]}{\overline{W}}. \quad (17)$$

The second term includes changes between the parental coordinate and its descendant mean under that attribution. It cannot be discarded merely because C1 identifies physical and experiential organization. Equation (17) is accounting, not a closed prediction until weights, descendant changes, and sampling are specified. Under C1, using the same coordinate on physical and experiential descriptions gives the same numerical identity; no additional phenomenal force is introduced.

6.4 Transmission and the limits of capability-only prediction

Faithful copying is not needed to formulate the more general question: can the distribution of present capability determine the next distribution? Let a finite represented state space have a fixed row-stochastic transmission kernel $K(s, s')$. With strictly positive capability-dependent weights $f(J(s))$, define

$$p'(s') = \frac{\sum_s p(s) f(J(s)) K(s, s')}{\sum_s p(s) f(J(s))}. \quad (18)$$

The kernel specifies descendant-state probabilities under a fixed model and context. It can represent transmission changes rather than exact copying, but it must be supplied independently of C1. Dependence on unrepresented history, population composition, or ecological feedback would require enlarging the state or modifying the model. The formula describes a deterministic distribution update; finite-population sampling is a further layer.

Write $q(j) = \sum_{s: J(s)=j} p(s)$. Say capability dynamics are *closed* if the same q always gives the same q' , for every starting distribution p in this model.

Proposition 6 - Capability closure under transmission. Under (18), closure holds if and only if, for every destination class j' ,

$$\sum_{s': J(s')=j'} K(s, s') = \bar{K}(j, j') \quad \text{for all } s \text{ with } J(s) = j. \quad (19)$$

When this condition holds,

$$q'(j') = \frac{\sum_j q(j) f(j) \bar{K}(j, j')}{\sum_j q(j) f(j)}. \quad (20)$$

Proof. If (19) holds, sum (18) over a destination class and group parental terms by j , giving (20). Conversely, choose point-mass initial distributions at two states in the same class. Both give the same initial q ; their strictly positive reproduction weights cancel on normalization. If their destination-class probabilities differ, their q' differ, contradicting closure. Thus those probabilities must be equal within every class. $\square$

Equation (19) is the strong-lumpability row-sum criterion (Tian & Kannan, 2006, Theorem 2.3), and the one-parent form of transmission ambivalence in evolutionary coarse-graining (Burjorjee & Pollack, 2006). Burjorjee (2007) considers restricted population domains and fitness means. Failure rules out closure for all starting distributions, not restricted or approximate closure. An equation whose coefficients still depend on hidden within-class composition is not autonomous at the J level. Section 7.2 supplies a circuit witness; its architecture-level model does not identify complete experience or actualize hypothetical descendants.

Conversely, let distinct complete types share one J -value, take $W=1$ and K to be the identity matrix. Capability dynamics close exactly although J does not identify experiential type. Thus dynamical sufficiency is not experiential identification. Noninjectivity alone implies neither future relevance nor closure failure. Adding mechanism coordinates may help prediction, but their own closure conditions must be checked.

7. Retention, usable information, and adaptive value

7.1 Memory without a larger sensory alphabet

For independent fair binary inputs, a current-input-only agent recalls the preceding bit with accuracy $1/2$, whereas reliable one-bit memory permits accuracy one. The sensory alphabet is unchanged. This is temporal reorganization, not an increase in every coordinate. In a separate faithful-inheritance model, let $m = 0, 1$ label the memoryless and memory architectures and assume positive reproductive weights

$$W_m = 1 + \beta J_m - cm, \quad \beta > 0, \quad c \geq 0.$$

Here $J_0 = 1/2$, $J_1 = 1$, β is the assumed benefit per unit success, and c the assumed cost. Memory is favored exactly when $c < \beta/2$, neutral at equality, and disfavored above it. These supplied costs and benefits concern inherited architecture classes, not complete-type copying or quantities derived from C1. For comparable actual realizations, Proposition 1 still applies to the capability difference.

7.2 Matched storage, identical current behavior, different accessible gains

A stronger control gives both systems the same one-bit storage capacity. Let independent fair bits X_t, N_t enter a clocked circuit with configuration bits w, r . Its memory and output obey

$$M_{t+1} = X_t \oplus (w \wedge N_t), \quad Y_t = \begin{cases} X_t, & r = 0, \\ M_t, & r = 1. \end{cases} \quad (21)$$

Here $\oplus$ denotes XOR. Output is emitted before the memory update. The declared task is always to output X_{t-1} , after a warm-up step. Circuit A has $w = 0$; circuit B has $w = 1$. They can use the same ideal gate/register layout, with different configuration values. This is an executable circuit model, not a measured neural system or a physically fabricated device.

Initially both have $r = 0$, so their outputs are identical for every shared input trace: $Y_t = X_t$. Both score $J = 1/2$. The memory marginal is a fair bit in either circuit, but A stores the preceding task input, whereas B stores that bit XOR an independent fair mask. Thus

$$H(M_t) = 1 \text{ bit in both}, \quad I(M_t; X_{t-1}) = \begin{cases} 1, & A, \\ 0, & B. \end{cases} \quad (22)$$

Changing the same readout bit to $r = 1$ raises A's score to one and leaves B's score at one half. B lacks a retained copy of the mask or another accessible previous-input channel. No task, input distribution, or scoring rule has changed.

Configuration	Stored bit	Output	Fixed-task accuracy
A0: $w = 0, r = 0$	Previous X	Current X	1/2
B0: $w = 1, r = 0$	Previous X XOR previous N	Current X	1/2
A1: $w = 0, r = 1$	Previous X	Memory	1
B1: $w = 1, r = 1$	Previous X XOR previous N	Memory	1/2

Now stipulate that a descendant inherits w , while $r = 0$ changes to 1 with probability μ , with no reverse change in this model. From A0, next-generation mean capability is $1/2 + \mu/2$; from B0 it remains 1/2. The transmission probabilities into the $J = 1$ class are μ and zero, violating (19) when $\mu > 0$. Reproductive weights depending only on the common parental score cancel at this step. Subsequent selection is separate and depends on net fitness.

The mechanism is task-relevant retention, not a larger register or higher marginal entropy. It illustrates a difference hidden by a *specified* current capability measure; a battery that tests the memory readout directly would distinguish the architectures already. Configuration inheritance is a stipulated population model, distinct from individual learning. Equal circuit size also does not establish equal energy cost.

Within UCT, experiential conclusions concern actual valid process realizations whose complete organization preserves the relevant distinction under the declared comparison. Circuit labels and coarse truth tables are not themselves complete experiential types. The example supports no scalar claim that A has more total experience, and the memory-information value in (22) is not a consciousness measure. Appendix C provides the noise boundary, transmission specification, and reproducible checks.

7.3 When retained information has adaptive value

The preceding recall task rewards retention directly. An ecological task need not. To make the distinction precise, fix a finite joint law of an environmental target Z , current observation X , and additional retained information M , a finite nonempty action set $\mathcal{A}$, and bounded payoff $u(z, a)$. A baseline policy observes X ; an enlarged policy observes (X, M) . A baseline policy may be any map from X to the common action set; an enlarged policy may be any map from (X, M) to that same set. No additional constraints couple decisions across observations. Both optimize the same objective, and the enlarged class includes every baseline policy by ignoring M . Define

$$V_X = \mathbb{E}_X \max_{a \in \mathcal{A}} \mathbb{E}[u(Z, a) \mid X], \quad V_{XM} = \mathbb{E}_{X, M} \max_{a \in \mathcal{A}} \mathbb{E}[u(Z, a) \mid X, M]. \quad (23)$$

Proposition 7 - Decision value of added information. Under these conditions, $\Delta V = V_{XM} - V_X \geq 0$. Equality holds if and only if, for every positive-probability x , there is at least one action optimal for every positive-probability m conditional on that x :

$$\bigcap_{m: \Pr(m|x) > 0} \arg \max_{a \in \mathcal{A}} \mathbb{E}[u(Z, a) \mid x, m] \neq \emptyset. \quad (24)$$

Proof. Fix x and take a baseline-optimal action a_x . The conditional improvement equals

$$\sum_m p(m \mid x) \left[\max_a \mathbb{E}[u(Z, a) \mid x, m] - \mathbb{E}[u(Z, a_x) \mid x, m] \right]. \quad (25)$$

Each bracket is nonnegative. Equality holds precisely when a_x is optimal for every supported m . Conversely, any action in the intersection in (24) attains the average of the conditional optima and is baseline-optimal. Averaging over supported x proves both claims. $\square$

This is a standard finite value-of-information argument, included to make the biological application explicit, not claimed as new decision theory (see Brooks, Frankel, & Kamenica, 2024). Conditional irrelevance, $Z \perp M \mid X$, suffices for zero gain, but is not necessary: different posteriors can favor the same action. Information can therefore refine a representation without improving this decision. The proposition concerns attainable optimal performance in nested policy classes. A real organism may fail to exploit the information, and acquiring memory may change its resource constraints, invalidating nesting. Under constrained but nested policy classes, weak improvement still follows from inclusion; the pointwise formulas and common-optimum equality criterion need not apply.

For an exact example, let Z_t be a fair stationary binary environment with symmetric transition probability $\Pr(Z_t = Z_{t-1}) = \alpha$. There is no informative current cue. A retained bit is $M_t = Z_{t-1} \oplus E_t$, where independent noise has probability $\epsilon \in [0, 1/2]$. The action predicts Z_t and payoff is accuracy. The baseline optimum is $1/2$, whereas

$$V_M = \frac{1}{2} + |2\alpha - 1| \left(\frac{1}{2} - \epsilon \right), \quad \Delta V = |2\alpha - 1| \left(\frac{1}{2} - \epsilon \right). \quad (26)$$

Indeed, the probability that memory matches the current state is $q = \alpha(1-\epsilon) + (1-\alpha)\epsilon$. Symmetry makes the optimal accuracy $\max(q, 1-q)$, yielding (26). Persistent environments favor using the retained bit; alternating environments favor its inversion. At $\alpha = 1/2$, even perfect memory of the past gives no benefit for this current-state task. This does not contradict Section 7.1, whose target

is the past bit itself. A sweep of α changes the evaluation environment and must not be presented as a fixed-J demonstration of NESIG.

For a separate two-architecture selection-only model with faithful architecture inheritance, stipulate positive weights $W_0 = 1 + \beta V_X$ and $W_1 = 1 + \beta V_{XM} - c$, with $\beta > 0$, $c \geq 0$. Starting with both architectures present, the memory architecture increases its deterministic frequency exactly when

$$\beta \Delta V > c. \quad (27)$$

Equality gives neutrality in this model. The fitness map and cost are supplied ecological assumptions, not consequences of C1; these architecture classes are not automatically complete process types. Thus three implications must remain separate: added information can improve achievable capability; realized improvement changes complete experiential type under the conditions of Proposition 1; greater capability yields a selection advantage only through the actual benefit-cost relation. No step ranks total experiential richness. Exact rational checks are reported in Appendix D.

8. Continuity, diversity, and realized repertoires

8.1 Continuity without a universal smooth trajectory

UCT I transfers continuity when physical and experiential descriptions share a declared structural space and topology or pseudometric. Evolutionary histories can branch, terminate, and change the appropriate descriptive dimension; they need not form one smooth curve. Imagining all ancestors clarifies the absence of an experience-onset gate within C1, but does not establish differentiability or an ordering of extant species. Smooth paths or arbitrarily fine chains require their own physical premises.

A lineage relation is also not identity of one subject persisting from prebiotic chemistry into a human. Organisms reproduce, develop, learn, and interact; actual process tokens may continue, transform, or end under UCT I. Genetic transmission, cultural transmission, individual learning, and ongoing neural dynamics therefore require separate state variables and update rules. The evolution of an architecture does not transfer an ancestor's complete experiential type into its descendants. What C1 preserves across these accounts is the identity of each actual organization with its own experiential organization, not a single ancestral stream.

The historical question concerns which organizations arose and why. Selection, transmission, drift, constraints, and ecological history contribute to that explanation; C1 supplies their experiential interpretation without adding a separate force.

8.2 Realized repertoires, coexistence, and frontiers

Let $\mathcal{A}_t$ be an explicitly selected ensemble of actual tokens at a time or across a declared interval. Define its realized repertoires:

$$\mathcal{R}_D(t) = \{s(P) : P \in \mathcal{A}_t\}, \quad \mathcal{R}_\Phi(t) = \{[\Phi(P)] : P \in \mathcal{A}_t\}. \quad (28)$$

Corollary 1 - Realized repertoire identity. Under C1,

$$\mathcal{R}_D(t) = \mathcal{R}_\Phi(t). \quad (29)$$

For any common, predeclared coordinate map $C : \mathfrak{S} \rightarrow \mathbb{R}^k$, their coordinate images and Pareto frontiers are therefore equal.

Proof. Equality holds element by element by (1). Applying the same coordinate map gives identical image sets; taking maximal elements under the same componentwise order gives identical frontiers. $\square$

Neither claim entails that the frontier increases over time. A contemporaneous repertoire can contract. A historical union can only gain members by definition, but this bookkeeping fact does not establish increasing complexity, better adaptation, or positive progress on every coordinate.

Abundance is a separate question. A toy population may retain one thousand tokens at coordinate (1, 1) while adding one at (3, 2). The modal coordinate remains (1, 1) while the Pareto frontier changes. UCT is compatible with this pattern because existence, abundance, and frontier position are different attributes. C1 alone does not predict the frequencies or explain their ecological causes.

The actual-token restriction also limits claims about evolvability. Identity on actual tokens does not automatically specify experiential types for every merely possible descendant architecture. A modal extension would need to be stated separately. Here, repertoire claims concern realized cases. Hypothetical architectures serve as conditional models, not as evidence that all mathematically described futures are physically accessible.

9. Competitor discrimination and theoretical limits

9.1 What would constitute a genuine formal disagreement?

Suppose a competitor T assigns complete phenomenal type by

$$\Phi_T(s) = H_T(A^*(s)), \quad (30)$$

where A^* is the complete descriptor that T actually regards as sufficient. If two admissible types satisfy $s \neq s'$ and $A^*(s) = A^*(s')$, UCT distinguishes their complete experiential types while (30) does not. This is a formal contrast if both statements concern the same token-level target and comparison domain.

UCT II's source-fidelity constraints prohibit substituting a scalar proxy for a competitor's complete descriptor. IIT 4.0's structured causal account and substrate commitments cannot be replaced by one integration score (Albantakis et al., 2023). Nor does one regional signal reproduce the broadcasting and self-monitoring distinctions of Dehaene, Lau, and Kouider (2017). Their broadcasting label "C1" is unrelated to UCT's C1. These are comparison requirements, not demonstrated refutations; NESIG may also hold in other structurally inclusive theories.

The following contrasts concern constitutive claims, not a completed empirical ranking:

Account	Relevant constitutive or methodological commitment	UCT's point of comparison
IIT 4.0 (Albantakis et al., 2023)	Experience is identified with structured causal distinctions and relations of a qualifying substrate, subject to exclusion	UCT admits valid overlapping or nested processes; it does not replace the complete IIT object with a scalar
Global availability and self-monitoring (Dehaene et al., 2017)	Information can become available for flexible use and be monitored; speech is not the entire mechanism	UCT treats these as substantive organizations without making either a universal existence gate
Higher-order approach (Brown, Lau, & LeDoux, 2019)	A suitable representation of a mental state plays a constitutive role; variants differ	UCT need not require that representation for every experience; absence of speech alone does not refute HOT
Quality-space comparisons (Fleming & Shea, 2024)	Relational quality structure can be investigated within several consciousness accounts	Similarity-based tests and multidimensional structure are shared ground, not exclusive UCT predictions

A bounded computation can also be physically unfolded across forward-connected stages while preserving its permitted input-output law. If the designated whole is an actual valid UCT token but fails IIT's whole-complex condition under a certified feed-forward substrate model, the assignments differ. Both accounts may nevertheless agree about all stipulated observations. Unfolding has established prior art (Doerig et al., 2019); neither a software graph nor hardware reuse alone certifies the relevant IIT classification. This is a conditional attribution contrast, not a demonstrated report-based refutation.

The non-exclusivity can be stated exactly. For any rival assignment Φ_T on the same domain, the full type-change implication $J(s) \neq J(s') \Rightarrow \Phi_T(s) \neq \Phi_T(s')$ holds if and only if J is constant on every Φ_T -fiber, equivalently $J = g \circ \Phi_T$ for some g on the realized image. This follows by the same representative-independence argument as Proposition 2. An injective phenomenal assignment is sufficient but unnecessary. A rival can retain everything relevant to J while merging other structural distinctions. Failure of the full implication requires a pair with unequal J and the same rival-assigned experience. A counterexample to improvement-only NESIG additionally requires that the pair constitute an improvement under the declared capability order. For a scalar capability with its usual total order, these conditions coincide.

The nearest prior accounts constrain the contribution more directly than a generic list of rival theories:

Prior account	Relevant overlap	Contribution of the present paper
Multidimensional animal consciousness (Birch et al., 2020)	Consciousness profiles need not lie on one scale	Specify exactly when a capability profile identifies a complete type or a selected coordinate
Phenomenal interface theory (Klein & Barron, 2025)	Architecture and the evolution of conscious organization	Analyze capability fibers under UCT's actual-token identity, without claiming architectural evolvability as a new idea
Evolutionary complexity account (Edelman, 2026)	Complexity and phenomenal differentiation through evolution	Separate type change, coordinate direction, population abundance, and frontier position

Prior account	Relevant overlap	Contribution of the present paper
Structure-quality selection framework (Williams, 2025)	Formal selection transfer, transmission, and representational relations	Distinguish complete-type identity from compression by capability, and ask when the compressed capability dynamics close
Evolutionary coarse-graining (Stephens & Waelbroeck, 1998; Burjorjee & Pollack, 2006; Burjorjee, 2007)	Projected selection/variation equations and conditions for closure	Apply existing criteria to capability classes; distinguish their physical predictive role from complete experiential identification
Neutral variation and evolvability (Wagner, 2008)	Phenotypically equivalent organizations can differ in accessible variation	Use a fixed-task, matched-storage circuit to separate capability, task-relevant retention, and the conditional experiential interpretation

Predictive-quotient accounts supply a closer formal comparison than a behavioral caricature. Appendix F retains the unsuccessful comparison with Rodgers’ author-hosted monograph (2026); its domain and realization conditions were not certified by the circuit example.

The table and this comparison identify overlap rather than certify originality. Price accounting, state aggregation, and unchanged physical predictions under an added aspect interpretation are not exclusive to UCT. The retained contribution is the conditional UCT interpretation and disciplined use of these distinctions. Their conjunction here does not establish historical priority for the underlying evolutionary framework.

9.2 Task coverage and the matched-circuit comparison

The factorization criterion has a useful operational specialization. Fix the comparison domain, task, preparation rule, labels, and physical context. Let a finite transcript τ retain every variable needed by a common score $h(\tau) \in [0, 1]$, and write $J(s) = \mathbb{E}_{P_s}[h]$. If equal rival-assigned phenomenal types imply equal laws P_s of this scored transcript, then J factors through the rival assignment and NESIG holds for this task. This is a sufficient condition: equal expectations do not require equal transcript laws. Across architectures, the correspondence must preserve the scoring labels and preparation rule; identical class names in different models are insufficient.

The approximate version is elementary:

$$|J(s) - J(s')| \leq \text{TV}(P_s, P_{s'}), \quad \text{TV}(P, Q) = \frac{1}{2} \sum_{\tau} |P(\tau) - Q(\tau)|.$$

To prove it, put $d = P_s - P_{s'}$. Since $\sum d = 0$, the positive and negative parts each have mass TV . For $0 \leq h \leq 1$, the sum $\sum hd$ lies between minus TV and TV . Exact equality of transcript laws gives the stated factorization. This standard probability bound is used as a comparison safeguard, not claimed as new mathematics or as an estimator from finite data.

The existing circuit exposes why transcript selection matters. After the readout edit, A1 emits X_{t-1} and B1 emits $X_{t-1} \oplus N_{t-1}$. Both output streams after warm-up are iid fair bits, although they are not identical on every shared input trace. For the fixed delayed-input task their accuracies are one and one half. The relevant transcript is (T, Y) , with $T = X_{t-1}$, rather than Y alone:

Scored pair (T,Y)	A1 probability	B1 probability
(0,0)	1/2	1/4
(0,1)	0	1/4
(1,0)	0	1/4
(1,1)	1/2	1/4

Output-only TV is zero; scored-joint TV and the accuracy gap are both one half. Output statistics therefore cannot substitute for a complete rival descriptor. Conversely, a descriptor preserving this scored joint law necessarily distinguishes the two task performances. No external target record is thereby declared part of experience or of a native grammar.

The output-law calculation establishes a comparison safeguard, not a completed phenomenal classification under a competing theory. Appendix F records the unfulfilled realization and descriptor requirements in the specific predictive-quotient comparison.

10. Empirical interfaces and discriminating predictions

The comparative argument must distinguish three forms of evidence: evidence for a physical mechanism, evidence for a finite report relationship, and evidence discriminating phenomenal assignments. Success at the first level does not automatically complete the other two. UCT I's finite operational correspondence packages already require a declared target and independently defended bridges (Liu, 2026a).

10.1 Residual organization beyond capability proxies

A finite empirical analysis can ask whether organizational features improve held-out prediction beyond task capability and theory-linked mechanism proxies. Appendix B separates additional target information from fitted-model error and repair of noisy measurements. A competitor's complete sufficient descriptor must not be replaced by a weak proxy and then declared insufficient merely because another measurement helps. The target, descriptor, comparison domain, and nuisance bounds must be fixed before interpreting a positive gain.

The COGITATE sample route remains prospective. No sample analysis or experimental confirmation of C1 is reported in this paper. A future analysis requires authorized data, inventory of existing derivatives, frozen trial-level features and a data dictionary, subject-grouped evaluation, and a defensible conditional null. The present article's conclusions do not depend on an unperformed analysis.

10.2 A limited internal-similarity candidate

For an accessible artificial model, fix a finite set Q of diagnostic continuations and response laws K_i^q for prepared states i . Use a common finite output alphabet and fixed nonnegative weights summing to one. The resulting weighted average of total-variation distances is a pseudometric on prepared-state labels: distinct states can have the same measured profile. Its relation to a model's comparison of its own processing episodes is an additional hypothesis, not a consequence of C1. Diagnostic forks must not feed their results into the report branch, and the actual process must retain the information needed for the comparison.

A fair binary randomization of which internal condition is nearer a reference, independent of every public record available to the observer, gives a precise public-input-only null: forced binary choice

has expected accuracy one half if the public record carries no information about that assignment. Success can establish private-information use under the protocol. A direct decoder remains an alternative to second-order monitoring, and neither mechanism by itself establishes felt similarity. Coordinate-invariant descriptors prevent one avoidable artifact: a description-only rescaling of an activation chart can reverse raw Euclidean rankings when Euclidean weights are retained, even though the process is unchanged and every physical map is transported consistently. Physically rescaling activations inside a fixed network is a different operation. These invariance requirements do not identify a unique experiential metric.

Similarity-based approaches have substantial antecedents in representational similarity analysis and quality-space research (Kriegeskorte, Mur, & Bandettini, 2008; Fleming & Shea, 2024). The candidate therefore proposes a scoped operational bridge, not an exclusive UCT test. A defensible next study must specify model, episode boundaries, access mechanism, calibration, negative controls, and held-out evaluation before making a stronger claim.

11. Discussion: what a unified explanation adds

The cross-domain account has one constitutive rule but several mechanisms of change. A mountain’s material history, a cell’s regulated response, a multicellular division of labor, an animal’s temporal control, a human’s conceptual report, and an artificial agent’s learned computation differ in organization. Within UCT, those differences are not incidental accompaniments of an otherwise undifferentiated experience. The relations actually constituting each process belong to its experiential organization. The biological or engineering account explains how the relations arise, what they enable, and why they persist.

This answers the mountain-cell contrast at the appropriate level. Neither bulk matter nor the label ‘alive’ explains adaptive control. Specific couplings among environmental sensitivity, retained state, action, and maintenance do. The comparison is not exhausted by a binary experience judgment because UCT assigns experience to both valid processes while distinguishing their organizations. It likewise avoids treating a fluent artificial narrator as automatically more experientially organized than an unreporting but structured computation.

The formal results clarify what is compressed by a capability score. Three questions must be kept apart: does J identify complete type, does fitness factor through J , and do transmission dynamics close on J ? They are not logically independent in every domain: injective J makes the first two trivial and its classes singleton. Their separation matters when J is noninjective. Current task equivalence can then hide different retained information and future opportunities, while exact capability-level prediction can coexist with incomplete experiential identification.

The identity interpretation does not make the organization causally idle. The same retention, comparison, and control relations that the physical model uses are identified as experiential organization; the theory does not posit an extra experience variable competing with them for causal influence. This is a constitutive claim, not an additional mechanism or an argument that every phenomenally relevant property is directly accessible to a reporter.

The constitutive identity is inherited, not established by these proofs or biological examples. Removing its experiential interpretation leaves the physical models and their empirical predictions unchanged. A discriminating consciousness test therefore needs a finite phenomenal target and a justified bridge to observation, as Section 10 explains. The present contribution is the comparative account and its exact conditions of application; physical predictive success alone does not independently validate C1.

12. Conclusion

From inorganic processes to artificial agents, UCT treats experience as actual organization and intelligence as capability under specified conditions. New forms of retention, coordination, monitoring, and expression can transform what a process does or reports while leaving other capability coordinates unchanged. A genuine capability change entails complete experiential-type change under a fixed evaluation; it does not entail a universal increase in experiential complexity. Evolutionary and engineered histories can produce different combinations of capability and reporting access because useful information, its exploitation, its cost, and its transmission are separate determinants. Self-report is one organizational achievement within this account, not the origin or exhaustive measure of experience. The resulting theory is a comparative and conditional synthesis whose next empirical task is a justified finite bridge and a source-faithful competing prediction.

Appendix A. Premise and result ledger

Result	Required premises	Excluded inference
Proposition 1: refinement/NESIG	C1; comparable complete types; fixed invariant capability map	A sampled score increase proves increased phenomenal richness
Strict refinement	Noninjective capability map on the declared domain	Every scalar or finite test must have a nontrivial fiber
Proposition 2: coordinate recovery	Constancy on each observation fiber	The inverse is automatically measurable, stable, or computable
Local fiber dimension	A justified smooth chart and locally constant rank for each map	Pointwise rank alone gives the dimension at singularities
Proposition 3: nonmonotonicity	A countermodel satisfying the stated premises	No restricted monotonic law could ever hold
Proposition 4: local selection null	Differentiable specified component that factors through intelligence	Finite neutrality or neutrality of total fitness
Proposition 5: exact class preservation	Deterministic selection-only reweighting; faithful copying; class-constant nonnegative weights; surviving class	Mutation, drift, or coarse-trait copying preserves complete-type conditionals
Proposition 6: capability dynamics closure	Finite state domain; fixed row-stochastic kernel; strictly positive class-constant reproductive weights; all initial distributions	Noninjective capability always gives unclosed dynamics, or closure must fail for every restricted population
Corollary 2: sensory partition transfer	Actual comparison cases; full-type equality and distinction	Equal behavior establishes equal experience
Proposition 7: added-information value	Finite fixed decision problem; bounded payoff; unrestricted maps to a common action set; no cross-observation constraints	Information necessarily improves a fixed task, or optimal capability automatically raises fitness
Corollary 1: repertoire/frontier identity	Same actual ensemble, coordinates, and order	Evolution necessarily expands a complexity frontier
Competitor contrast	Source-faithful sufficient descriptor; same target and domain	A difference from an analyst's proxy refutes the source theory
Task-coverage implication	Same scored transcript, evaluation and preparation; rival equality preserves its law	Output-only equality proves phenomenal equality, or this implication is necessary
Residual prediction bound	Finite latent descriptor and expected fitted losses; specified target bridge; independently bounded nuisance terms	Any held-out gain uniquely confirms UCT

Two short boundary examples are especially useful. First, $J(x) = x^3$ is injective, although its derivative at zero has rank zero. Its zero-fiber is a singleton, so a pointwise substitution into $n - r$ is invalid there. Second, additional information need not improve every prediction loss: if a fair bit R selects between $Y = 0$ and an equal mixture of $Y = -1, +1$, both conditional means are zero. Knowledge of R changes the conditional distribution but does not improve optimal squared-error prediction. These examples delimit regularity and target assumptions rather than challenge C1.

Appendix B. Residual information and fitted prediction

Let Z include baseline measurements and controls, R additional organization features, and Y a finite target. The term residual describes their proposed role; it does not establish statistical independence or orthogonality. For discrete Y and finite entropies, the optimal log-loss improvement is

$$H(Y | Z) - H(Y | Z, R) = I(Y; R | Z). \quad (31)$$

For fitted predictors, assume both population expected log losses are finite under the evaluation law. In particular, assigning zero probability to a positive-probability target event is excluded. Finite Y alone does not ensure this integrability. With the subtraction well defined,

$$L(q_Z) - L(q_{ZR}) = I(Y; R | Z) + K_Z - K_{ZR}, \quad (32)$$

where each K is the corresponding population conditional-KL excess risk. A positive predictive gain can reflect a better approximation to information already available in the baseline. It is not automatically evidence of new target information, much less a measurement of complete experience.

If a competitor's sufficient descriptor A^* is measured incompletely, additional features can also repair that measurement. For a finite-valued A^* , define

$$u = H(A^* | Z), \quad d = I(Y; R | A^*, Z).$$

Then the conditional information chain rule gives

$$I(Y; R | Z) \leq u + d. \quad (33)$$

If baseline excess risk is independently bounded by η , (32) implies fitted population gain at most $u + d + \eta$. The derivation follows below. Phenomenal sufficiency alone does not ensure $d = 0$: the report mechanism or its noise may still depend on R . The finite-entropy bound cannot be transferred to a continuous descriptor by substituting differential entropy.

For finite A^* and quantities for which the following informations are defined,

$$\begin{aligned} I(Y; R | Z) &\leq I(Y; A^*, R | Z) \\ &= I(Y; A^* | Z) + I(Y; R | A^*, Z) \\ &\leq H(A^* | Z) + I(Y; R | A^*, Z). \end{aligned} \quad (34)$$

The first step adds nonnegative conditional information; the second is the chain rule; the third uses the entropy bound for a discrete descriptor. For a predictor q_Z , the expected log loss separates into the conditional entropy and conditional KL divergence:

$$L(q_Z) = H(Y | Z) + \mathbb{E}_Z D_{\text{KL}}(p(Y | Z) \| q_Z(Y | Z)). \quad (35)$$

Subtracting the analogous expression for (Z, R) yields (32). Combining it with (34) and $0 \leq K_{ZR}$, $K_Z \leq \eta$ gives the stated bound. These are population statements for fixed prediction rules, or conditional on their training data. Sampling uncertainty must be handled separately. For discrimination, u , d , and η must also be fixed or independently bounded before interpreting the held-out gain, including uncertainty in those bounds. An unconstrained bound cannot make a residual effect theory-specific.

A fair-bit example shows the bound's relevance. Let $Y = A^* = R$ and let Z be A^* corrupted by independent binary noise of probability 0.1. Then $d = 0$, yet

$$I(Y; R | Z) = H(A^* | Z) = h(0.1) \approx 0.325083 \text{ nats}. \quad (36)$$

A substantial residual effect can therefore coexist with complete sufficiency of the true descriptor. It arises because the residual repairs its noisy measurement. This constructed distribution is a methodological counterexample, not neural evidence.

Appendix C. Circuit protocol, transmission, and controls

Equation (21) describes two phase-ordered operations per input: emit the selected output using the old memory, then write the new memory. The first input is warm-up and is excluded from scoring. Both input streams are independent and identically distributed fair bits; the current input, previous mask, or an external recorder must not provide an undeclared delayed channel to the readout. The two configuration bits are fixed during each episode. The retained-memory register is one bit in every configuration; its entropy is evaluated across the declared input ensemble, not inferred from one episode.

For noise sensitivity, replace the update by

$$M_{t+1} = X_t \oplus (w \wedge N_t) \oplus E_t, \quad E_t \sim \text{Bernoulli}(\epsilon), \quad 0 \leq \epsilon \leq 1/2, \quad (37)$$

with independent noise. A1 has accuracy $1 - \epsilon$; the other three configurations have accuracy $1/2$. Both memory marginals remain uniform. A retains $1 - h_2(\epsilon)$ bits about the previous target, whereas B retains zero, where h_2 is binary entropy in bits. The mean gain from A0 under the stipulated readout mutation is

$$\mu(1/2 - \epsilon), \quad (38)$$

and is zero from B0. At $\epsilon = 1/2$, the fixed-task scores all coincide and this particular closure witness disappears. At $\mu = 0$, no initially unexposed readout is changed. These are explicit negative controls, not exceptions to NESIG.

The noiseless transmission matrix in state order A0, B0, A1, B1 is

$$K = \begin{pmatrix} 1-\mu & 0 & \mu & 0 \\ 0 & 1-\mu & 0 & \mu \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (39)$$

For $\mu = 1/10$, the offspring of A0 comprise one tenth A1 and nine tenths A0. Under a subsequent selection-only step with $W = 1 + J$ and no extra architecture-dependent cost, the A1 fraction becomes $4/31$. This fitness rule is supplied as an illustration, not measured or derived from equal gate counts. Introducing a different cost or transition rule can change the evolutionary outcome.

The circuit enumeration script `uct_iii_r14_circuit_model.py` exhausts all 1,024 pairs of five-bit input and mask traces and scores four post-warm-up steps per pair, giving 4,096 scored steps per configuration. It also computes exact weighted truth tables at $\epsilon = 0, 1/4, 1/2$, checks $\mu = 0, 1/10, 1/2, 1$, and records example internal/output traces. These finite computations check the model; the accuracy and information formulas follow from independence and the XOR update. Neither constitutes human, animal, or machine-phenomenology data.

The task-coverage audit, `uct_iii_r16_task_coverage_audit.py`, reuses the circuit function from `uct_iii_r14_circuit_model.py`. Exact enumeration over output horizons one through six covers 21,840 input/mask trace pairs and confirms equal post-edit output laws. The all-horizon conclusion follows analytically because successive outputs depend on independent input/mask pairs. Under the independent write noise in (37), output laws still coincide, while the single-trial scored-joint TV and accuracy gap both equal $1/2 - \epsilon$. Exact checks at $\epsilon = 0, 1/4, 1/2$ include all 16 indicator scores on the four-outcome joint alphabet. These checks concern observable distributions; they do not certify a competitor's physical realization or classify experience.

Appendix D. Exact decision and memory checks

The decision-value audit, `uct_iii_r18_memory_value_audit.py`, enumerates all 6,560 nonzero joint tables for binary (X, M, Z) with unnormalized weights in $\{0, 1, 2\}$. For binary prediction utility, it compares all four baseline and sixteen enlarged deterministic policies against the conditional-optimization formulas and the common-optimum criterion (24). All checks pass. There are 2,592 strict-gain tables and 3,968 equal-value tables; 3,008 tables have conditional statistical dependence between target and added signal despite zero decision gain. These are counts in the chosen finite grid, not estimates of biological prevalence. The general bounded-payoff result follows from the proof, not from this binary enumeration.

For (26), exact rational enumeration covers $\alpha \in \{0, 1/4, 1/2, 3/4, 1\}$ and $\epsilon \in \{0, 1/4, 1/2\}$, independently maximizing over the two baseline and four memory policies. All fifteen cases match the formula. Separate positive-weight checks at costs below, at, or above the possible benefit verify the frequency-update criterion, including zero-gain cases where no nonnegative cost is strictly below the benefit. Results retain optimal policies and the negative controls. They check stipulated models and contain no empirical measurements of organisms or experience.

Appendix E. Full-type distinctions and sensory coordinates

Let X index a declared family of actually instantiated stimulus-process cases. For architecture a , write $s_a(x)$ for the complete type of the corresponding actual token. Define

$$x \equiv_D^a x' \iff s_a(x) = s_a(x'), \quad x \equiv_\Phi^a x' \iff \Phi(s_a(x)) = \Phi(s_a(x')). \quad (40)$$

Corollary 2 - Transfer of actual structural distinctions. Under C1, $\equiv_D^a \equiv \equiv_\Phi^a$. Any strict refinement of these full-type physical partitions across two such instantiated families is the same strict refinement under the experiential description.

Proof. Substitute (1) into (40). $\square$

The full-type qualification prevents an important error. Equality of a behavioral response or receptor readout is not equality of complete structure. A coarse sensory partition may be useful scientifically, but transferring it into a claim about a particular phenomenal coordinate requires a declared bridge. This corollary alone does not identify what a stimulus feels like to another species or prove that an absence of verbal discrimination is an absence of structural differentiation.

For an illustrative representational coordinate, partition four inputs into $\{0\}$ and $\{1, 2, 3\}$ in architecture A, and into four singletons in architecture B. Both can perform the task “is the input nonzero?” perfectly. The additional distinctions refine this coordinate without improving the selected task. A claim about refinement of complete-type partitions additionally requires the relevant equalities as well as inequalities of complete actual types to be established. Complete episodes might already differ for unrelated reasons. The biological question is which sensory organization changes, rather than whether some physical difference exists anywhere in the episode.

Appendix F. Preserved unsuccessful predictive-quotient comparison

The retained comparison with Rodgers’ author-hosted monograph (2026, Chapters 16-18) concerns predictive sufficiency: it develops native predictive quotients, requires transition congruence, and separates phenomenal identification from the physical kernels. It also assigns a disjoint partition of qualifying subject cores. UCT instead starts from complete actual token organization and permits overlapping or nested processes. This is a difference of proposed constitution and domain, not a behavioral refutation. The following paragraph records why the matched circuits did not complete a source-faithful experiential disagreement.

For the Rodgers comparison, the relevant source commitments are explicit: A1 admits qualifying recurrent cores, and A2 identifies experience with their complete endogenous predictive object (2026, Chapter 17). Admission requires a supplied physical realization and executable return conditions (Chapter 15). The circuit specification in Section 7.2 supplies neither that certificate nor a source-faithful native-record assignment. Its readout edit is a stipulated transmission event, not an established native action. Moreover, ensemble output equality does not establish equality of pointed intrinsic-state objects. The present case therefore yields no completed rival phenomenal assignment, positive or negative. The failed discrimination attempt is a substantive limit on the example, while its information-retention and transmission results remain intact.

Declarations

Research status and availability. This theoretical preprint reports analytic arguments, stipulated examples, finite computational checks, and prospective methodology. It reports no newly acquired human or animal data and no new experiment on a pretrained language model. The supplementary reproducibility package accompanying this version contains the circuit, task-coverage,

and decision-value audit scripts and their exact-check outputs, together with a file manifest and reproduction instructions. Prospective COGITATE and model-report studies are not presented as completed experiments.

AI assistance. ChatGPT assisted with source retrieval, formalization, drafting, editing, computational checks, and critical review under the author’s direction. These activities are not independent peer review. The author is responsible for the manuscript and its claims. The source audit was targeted and does not establish exhaustive historical priority.

Relation to the Trinity Accord. This is adjacent, non-amending research. It does not define, amend, validate or authoritatively interpret the Trinity Accord or its Bitcoin Originals.

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